SOC AUTOMATION

A PROJECT REPORT

BY

Muhammed Razal TM

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Abstract

As cyber threats grow more complex, building effective Security Operations Centers (SOCs) is essential. This project focuses on creating an automated SOC home lab to help cybersecurity learners and professionals practice in a safe environment. The main goals include getting hands-on experience with open-source tools like Wazuh for alerts, Shuffle for automating responses, and TheHive for managing cases. The lab uses virtualization and cloud services to connect different security tools, making it easier to respond to incidents. By simulating cyber attacks and using tools like Sysmon to track activities, participants can learn to detect and respond to security issues. This SOC home lab is a valuable resource for improving skills and understanding in the fast-changing world of cybersecurity.

Introduction

In today's rapidly evolving cybersecurity landscape, organizations face increasingly sophisticated threats, necessitating the need for robust Security Operations Centers (SOCs). A SOC is crucial for detecting, analyzing, and responding to security incidents to safeguard critical assets and data. However, the process of building, configuring, and maintaining a SOC can be resource-intensive, often requiring significant expertise and infrastructure.

This project aims to build an Automated SOC Home Lab, drawing inspiration from the YouTube series by MyDFIR. The goal is to create a simulated SOC environment where cybersecurity enthusiasts and professionals can gain hands-on experience, familiarize themselves with real-world tools, and practice incident detection and response in a controlled environment. By leveraging open-source tools and cloud services, the project will provide an opportunity to explore SOC operations, incident response workflows, and integration of multiple security technologies.

Requirements

Infrastructure Requirements

Component	Details
Virtualization Platform	VirtualBox (Oracle)
Operating System	Ubuntu 24.04 LTS (VM for TheHive)
Endpoint	Windows 11 (Physical Machine for Wazuh Agent)
Wazuh Manager	Deployed server at 10.92.44.113

Software and Tools

Component	Details
Wazuh	Open-source SIEM and XDR platform for security monitoring and event management
Sysmon	Windows System Monitor — captures process creation, network connections, file modifications
TheHive	Open-source incident response and case management platform (v5.2.14)
Shuffle	Open-source SOAR platform for security orchestration and automation
VirusTotal	IOC enrichment — checks file hashes against 70+ antivirus engines
Cassandra	NoSQL database backend for TheHive data storage
Elasticsearch	Search and indexing engine for TheHive
ngrok	Secure tunnel to expose local TheHive instance to Shuffle cloud

VirtualBox: developed by Oracle, is a free, open-source virtualization software that lets users run multiple operating systems as virtual machines (VMs) on a single host. It is widely used for testing and development, allowing users to create isolated environments without needing dedicated hardware.

Wazuh: is an open-source platform that provides Security Information and Event Management (SIEM) and Extended Detection and Response (XDR) capabilities. It enables organizations to monitor their systems for security threats, analyze logs, and ensure compliance.

TheHive: is an open-source incident response and security incident management platform that works well with tools like Wazuh. It helps security teams organize, investigate, and track incidents by creating and managing cases.

Sysmon, or System Monitor: is a Windows tool that records detailed system activity to help detect malicious behavior. It captures events like process creation, network connections, and file modifications. Sysmon is commonly integrated with SIEM tools to enhance visibility into system events.

Shuffle: is an open-source Security Orchestration, Automation, and Response (SOAR) platform that enables security teams to automate tasks and create workflows. With a low-code interface, Shuffle allows users to integrate different security tools and automate response processes.

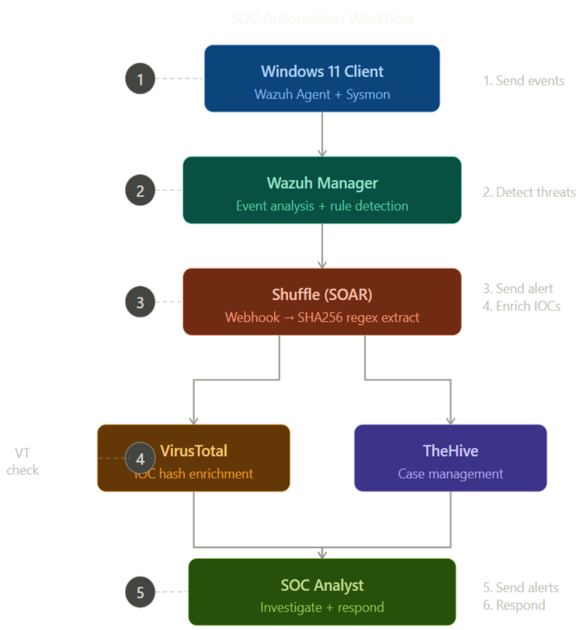
Proposed System

Workflow

The SOC Automation pipeline consists of the following steps:

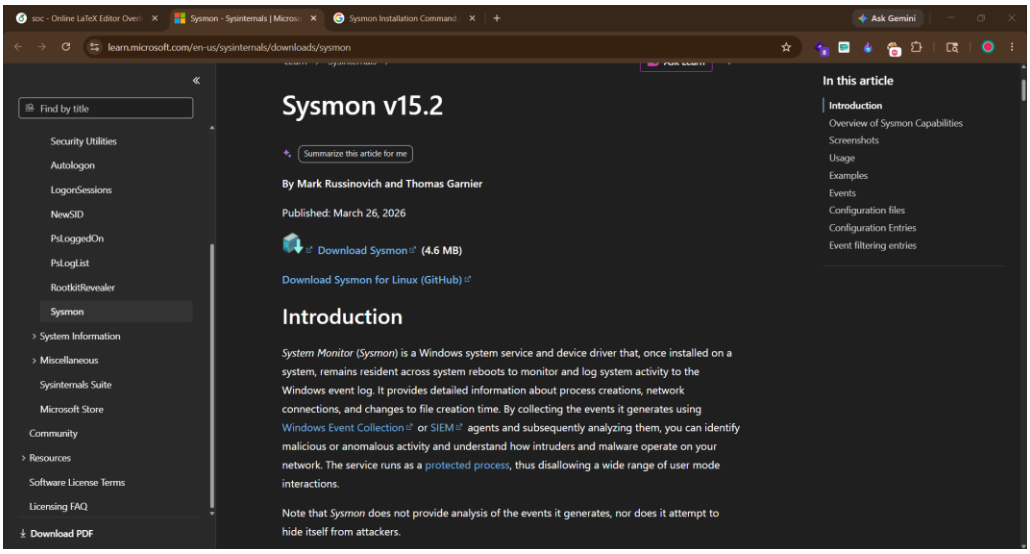
1. Windows 11 Client with Wazuh Agent (Event Collection): The agent gathers security-related data such as log entries, network traffic, and system modifications. This data is transmitted to the Wazuh Manager for centralized analysis.
2. Wazuh Manager (Event Aggregation and Analysis): Serves as the centralized processor, applying detection rules and algorithms to identify abnormal behaviors and generate alerts.
3. Alert Generation (Sending Alerts to Shuffle): When threats are flagged, Wazuh creates alerts and sends them to Shuffle via webhook integration.
4. IOC Enrichment in Shuffle: Alerts undergo enrichment where the SHA256 hash is extracted and cross-checked with VirusTotal to add contextual threat intelligence.
5. Sending Refined Alerts to TheHive: After enrichment, Shuffle creates a structured alert in TheHive for case management and investigation.
6. SOC Analyst (Incident Analysis and Response): The SOC Analyst investigates each incident using the enriched data and case history in TheHive.

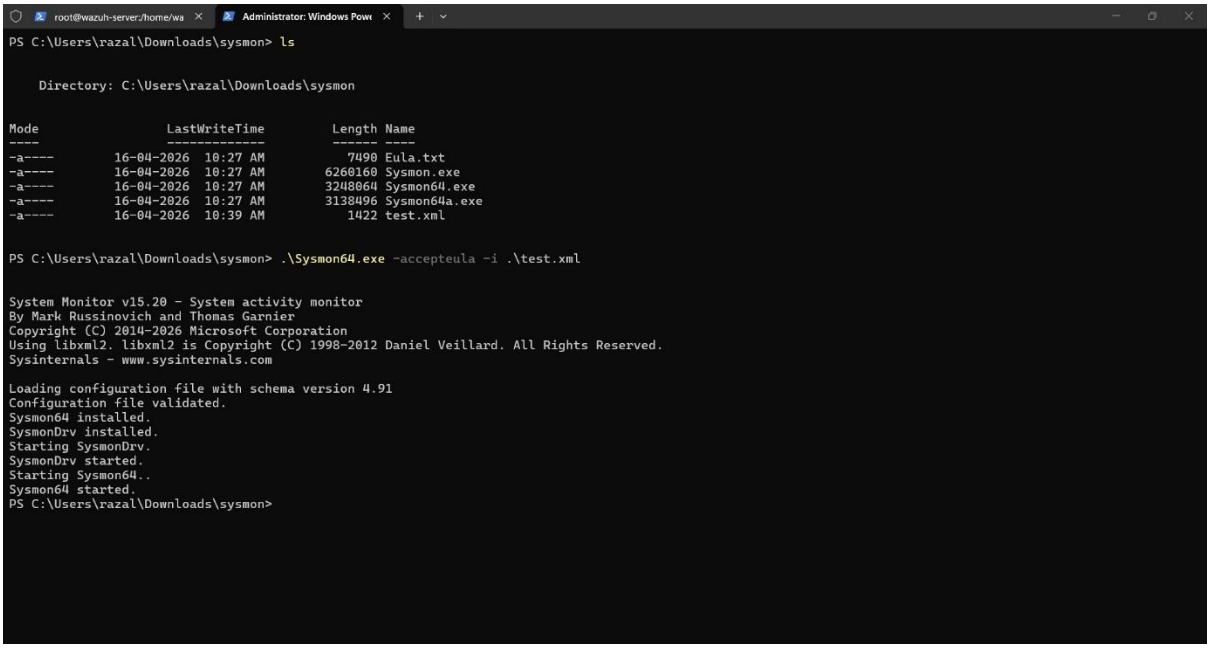
Workflow Diagram



Screenshots

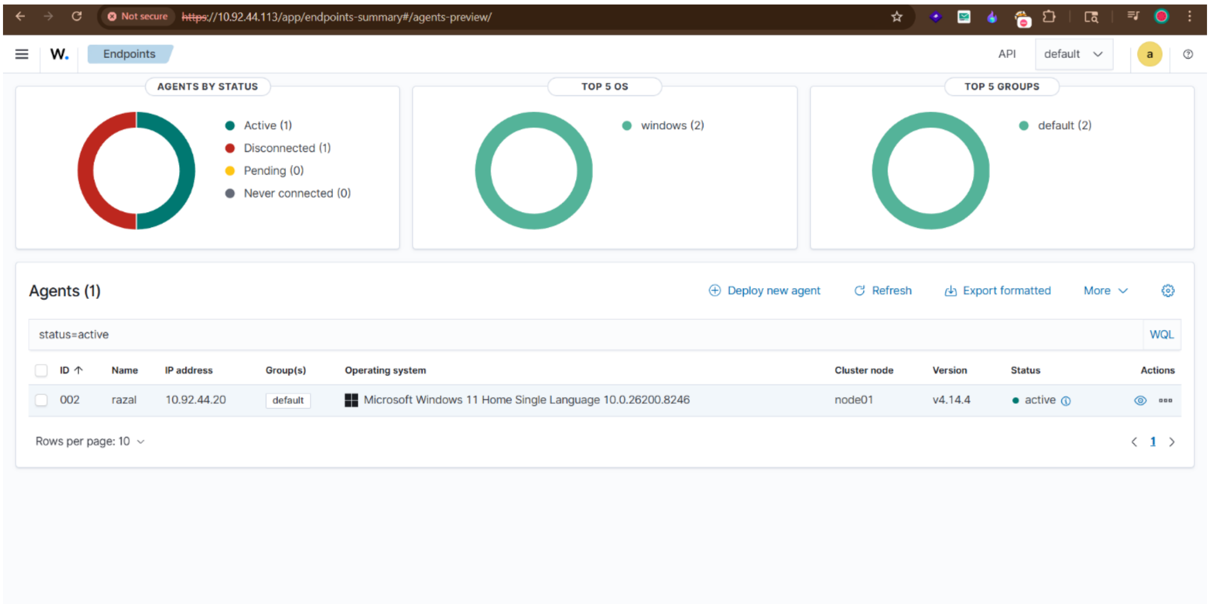
Downloading and Configuring Sysmon in Windows 11

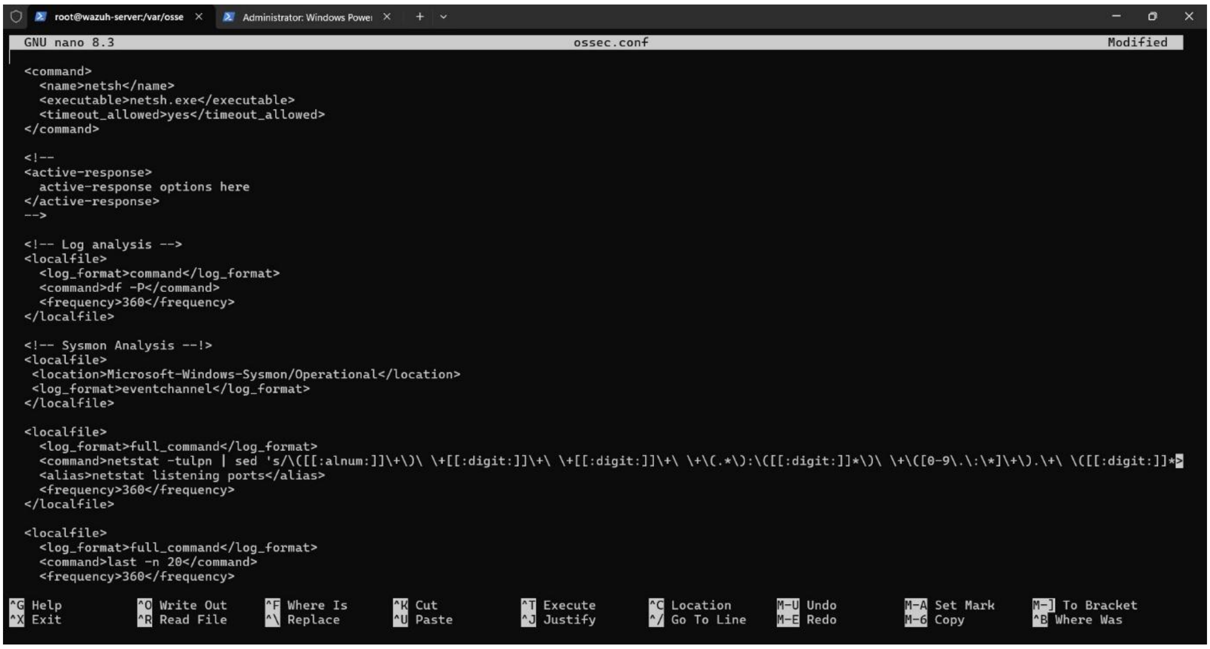




Setting up Wazuh Agent on Windows 11

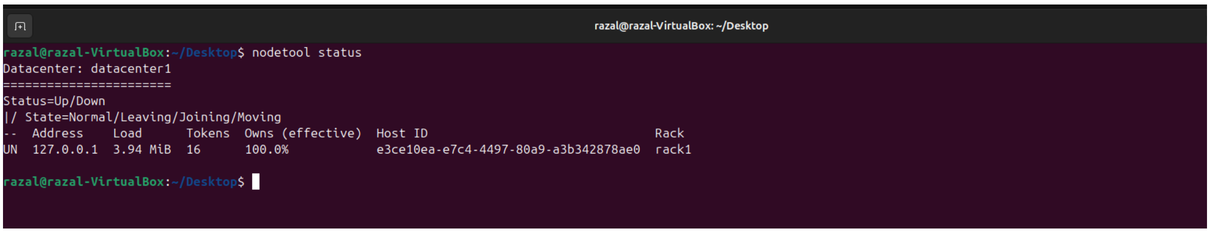
The Wazuh agent was installed on Windows 11 and configured to connect to the Wazuh Manager at 10.92.44.113. The ossec.conf was configured to forward Sysmon and Windows Security logs

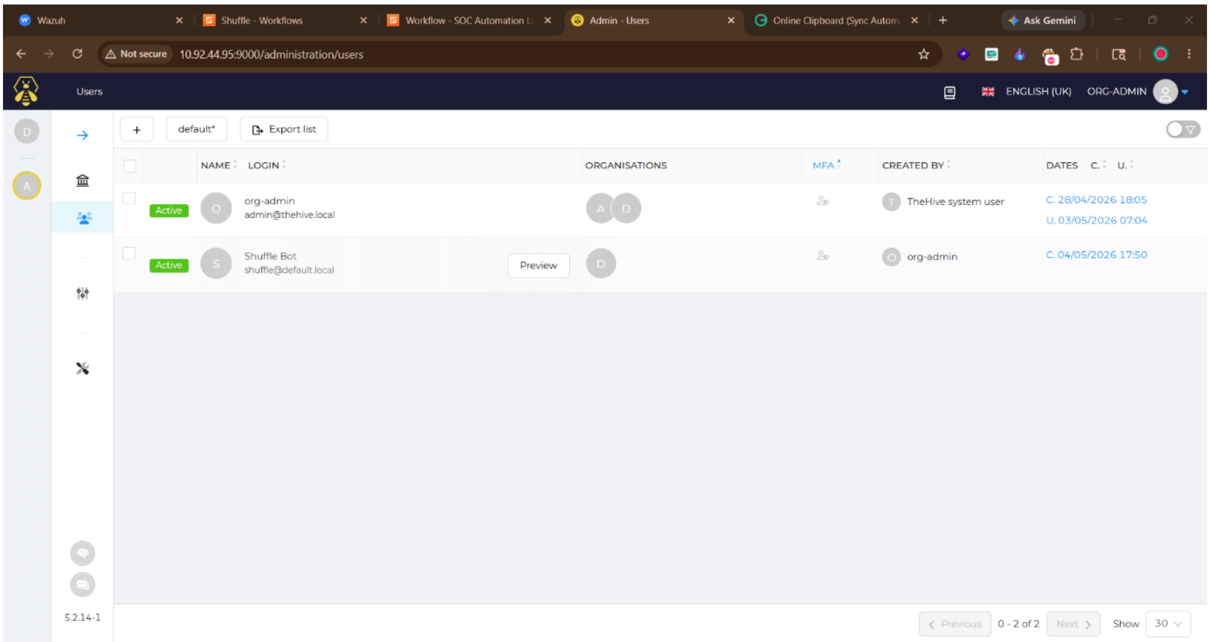




Installing TheHive on Ubuntu VM

TheHive 5.2.14 was installed on Ubuntu 24.04 in VirtualBox. Cassandra and Elasticsearch were configured as dependencies



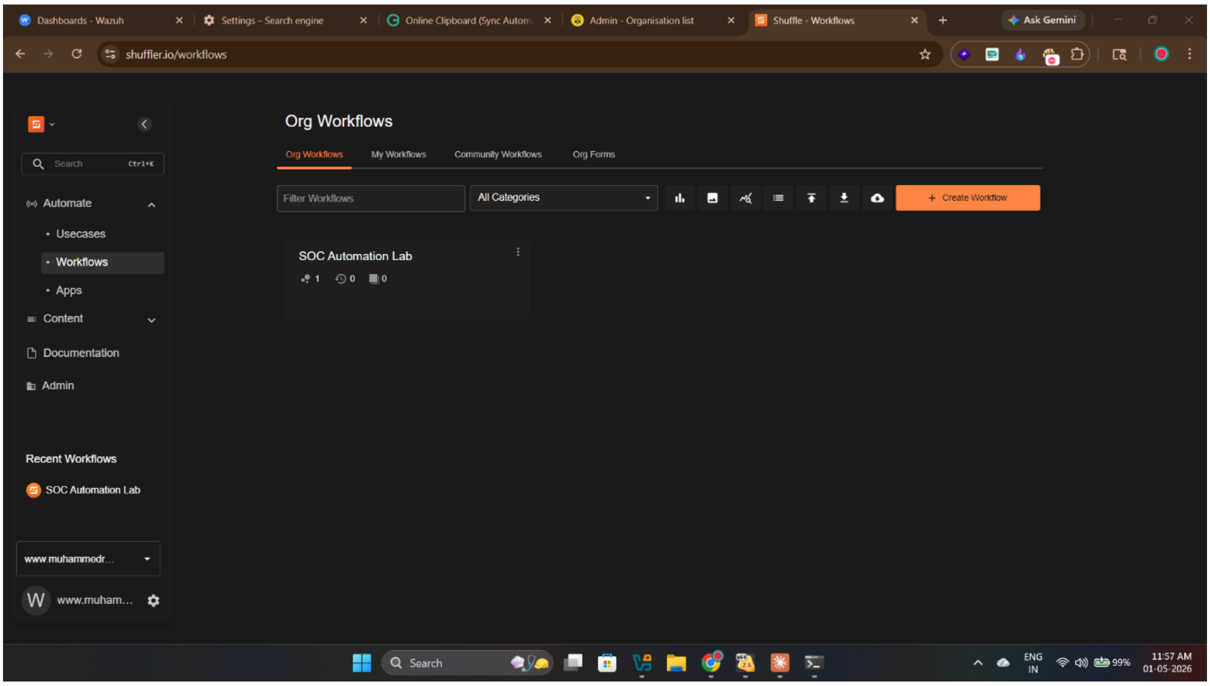

Wazuh
Shuffle - Workflows
Workflow - SOC Automation L
Admin - Users
Online Clipboard (Sync Autom
Ask Gemini
Not secure
10.92.44.95:9000/administration/users
Users
ENGLISH (UK)
ORG-ADMIN
default*
Export list
NAME
LOGIN
ORGANISATIONS
MFA
CREATED BY
DATES
C.
U.
Active
org-admin
admin@thehive.local
TheHive system user
C. 28/04/2026 18:05
U. 03/05/2026 07:04
Active
Shuffle Bot
shuffle@default.local
Preview
org-admin
C. 04/05/2026 17:50
5.2.14-1
Previous
0 - 2 of 2
Next
Show
30

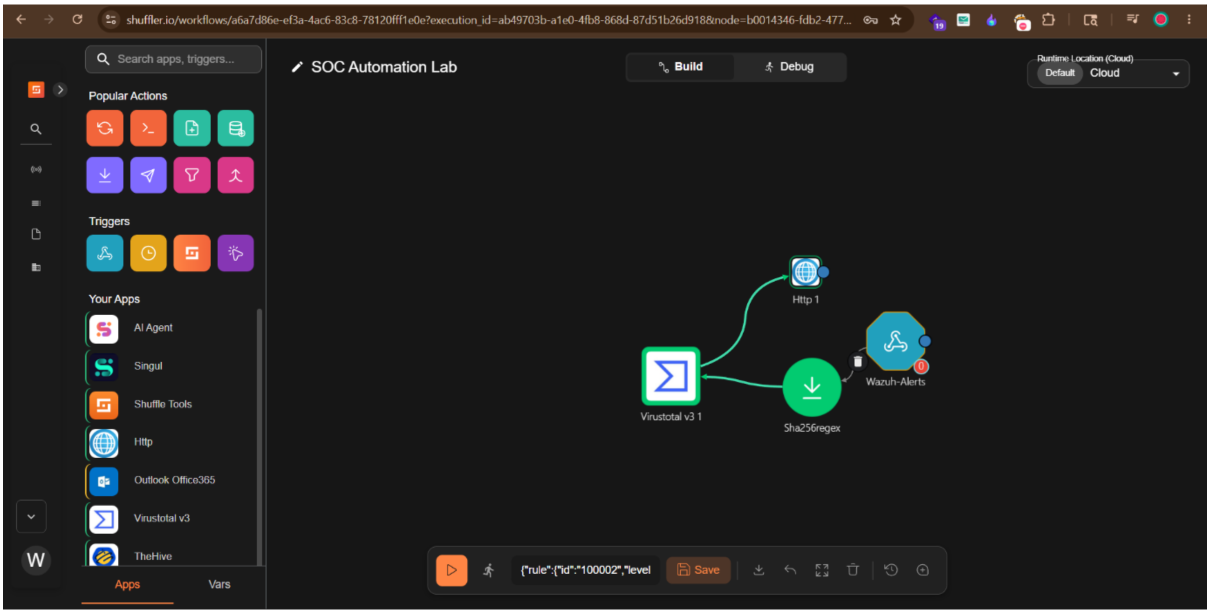
Creating Firewall Rules

UFW firewall rules were configured on the Ubuntu VM to allow inbound connections on port 9000 for TheHive.

Setting up Shuffle

A Shuffle account was created and the SOC Automation Lab workflow was built at shuffler.io.

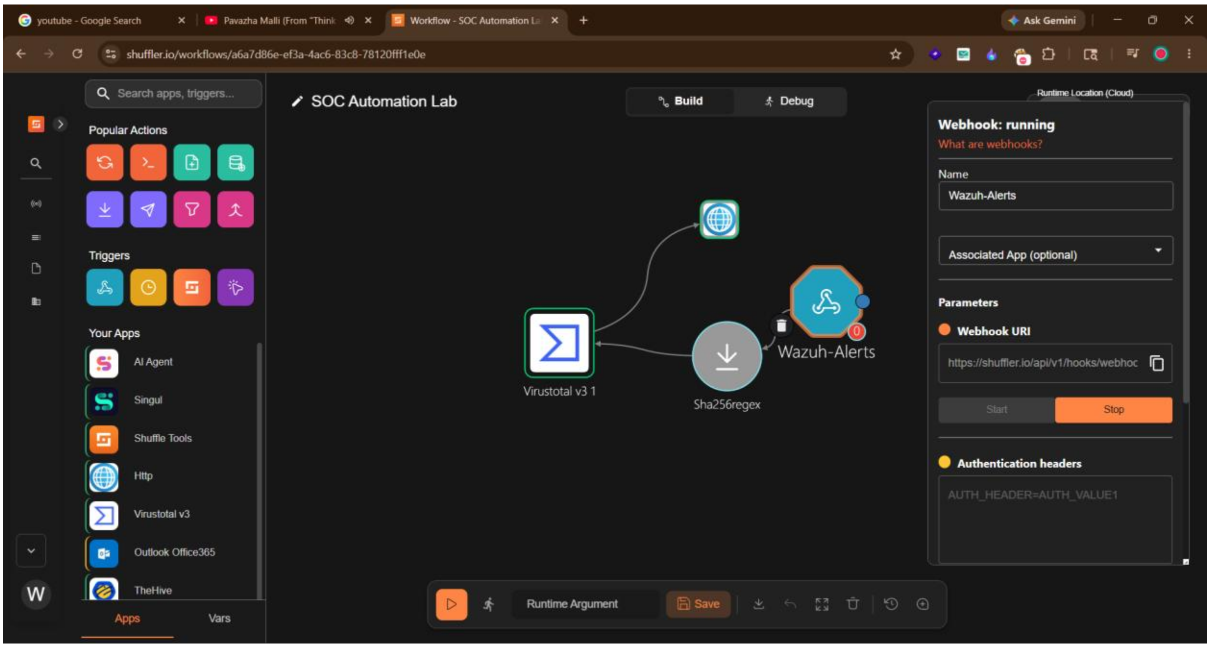




shuffler.io/workflows/a6a7d86e-ef3a-4ac6-83c8-78120fff1e0e?execution_id=ab49703b-a1e0-4fb8-868d-87d51b26d918&node=b0014346-fdb2-477...
Search apps, triggers...
SOC Automation Lab
Build
Debug
Runtime Location (Cloud)
Default
Cloud
Popular Actions
Triggers
Your Apps
AI Agent
Singul
Shuffle Tools
Http
Outlook Office365
Virustotal v3
TheHive
Apps
Vars
Http 1
Wazuh-Alerts
Virustotal v3 1
Sha256regex
{"rule":{"id":"100002","level
Save

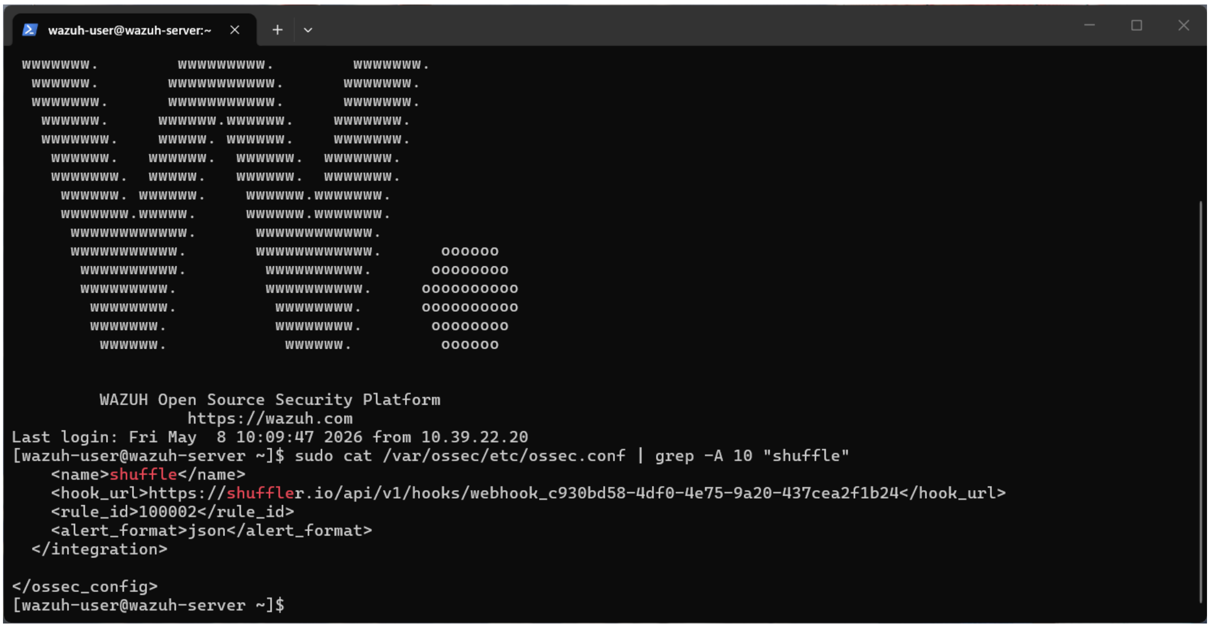
Adding Webhook Trigger

A Wazuh-Alerts webhook trigger was added to the workflow and started to receive alerts from Wazuh Manager



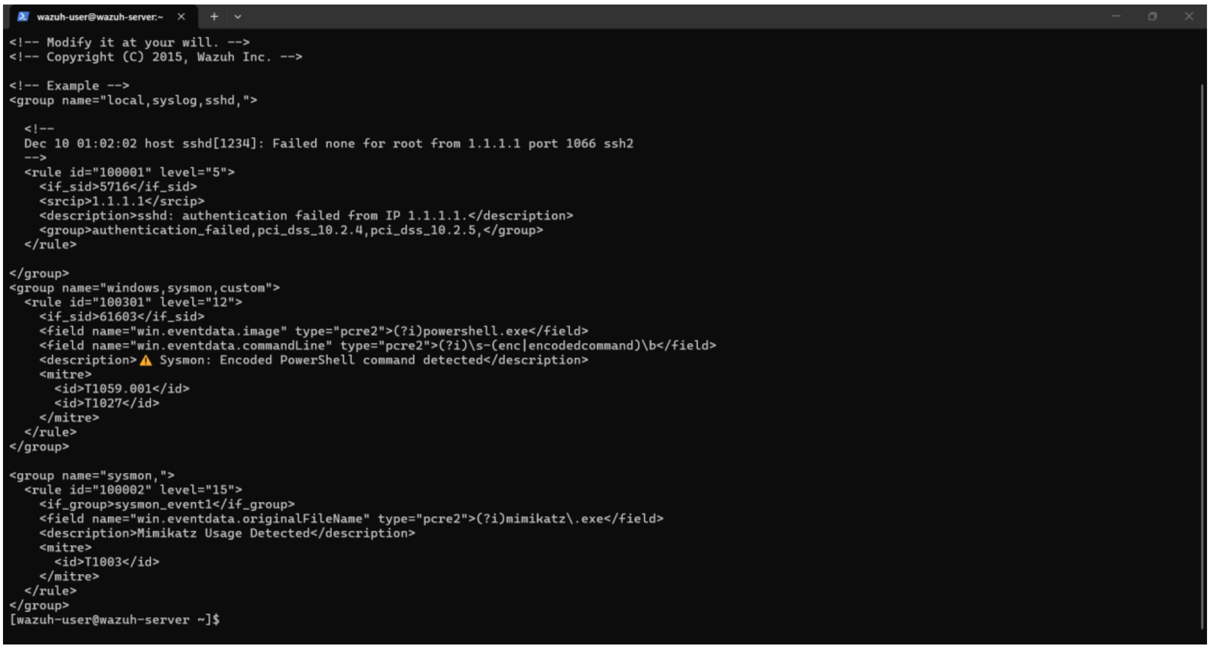
Configuring Wazuh to Connect to Shuffle

The ossec.conf file on the Wazuh Manager was modified to add the Shuffle integration block with the webhook URL and rule ID 100002



Creating Custom Mimikatz Detection Rule

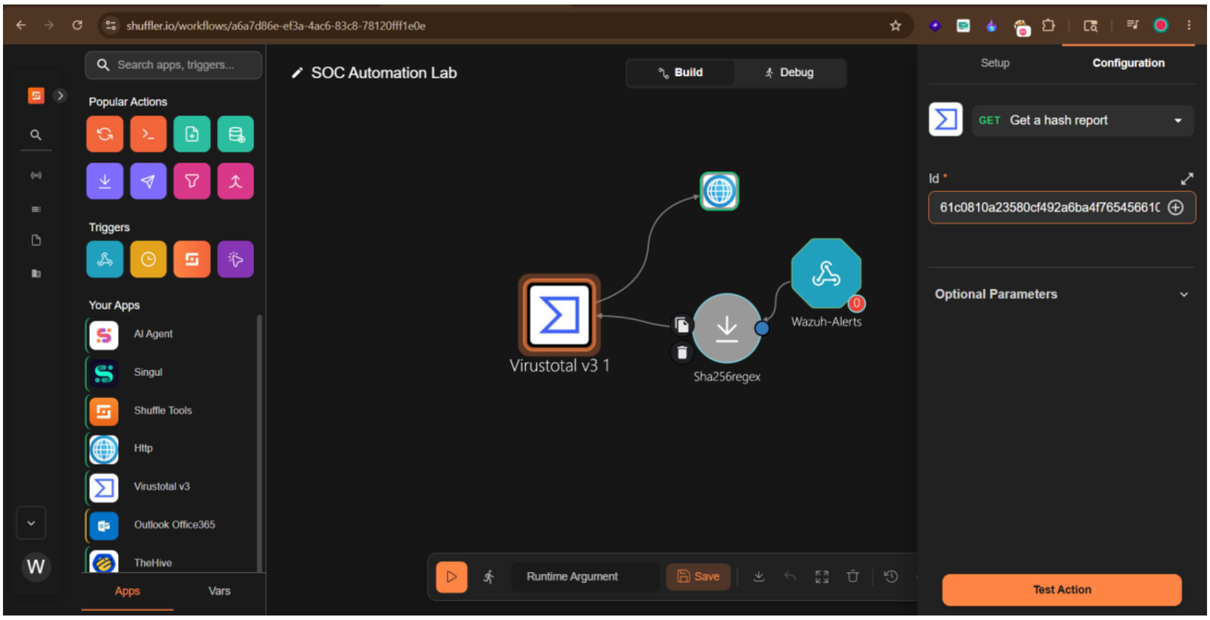
A custom Wazuh rule (ID 100002) was created in local_rules.xml to detect Mimikatz execution based on the originalFileName field.

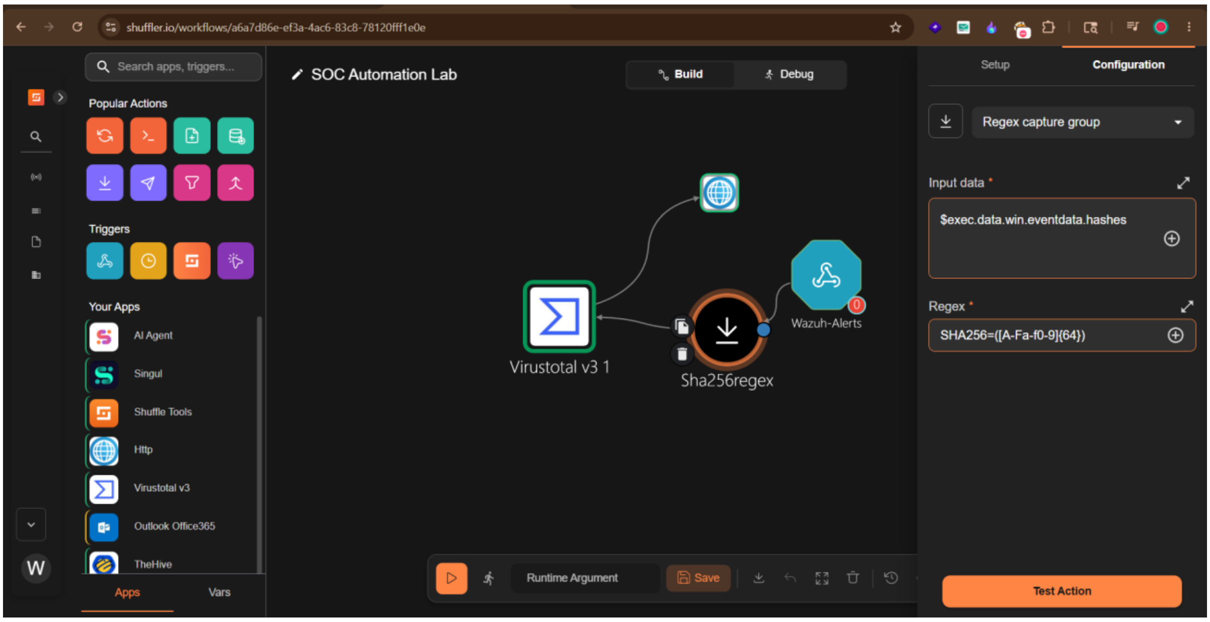


```
<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->

<!-- Example -->
<group name="local,syslog,sshd,">

  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>

</group>
<group name="windows,sysmon,custom">
  <rule id="100301" level="12">
    <if_sid>61603</if_sid>
    <field name="win.eventdata.image" type="pcre2">(?i)powershell.exe</field>
    <field name="win.eventdata.commandLine" type="pcre2">(?i)\s-(enc|encodedcommand)\b</field>
    <description>⚠ Sysmon: Encoded PowerShell command detected</description>
    <mitre>
      <id>T1059.001</id>
      <id>T1027</id>
    </mitre>
  </rule>
</group>

<group name="sysmon,">
  <rule id="100002" level="15">
    <if_group>sysmon_event1</if_group>
    <field name="win.eventdata.originalFileName" type="pcre2">(?i)mimikatz\.exe</field>
    <description>Mimikatz Usage Detected</description>
    <mitre>
      <id>T1003</id>
    </mitre>
  </rule>
</group>
[wazuh-user@wazuh-server ~]$
```

Integrating VirusTotal

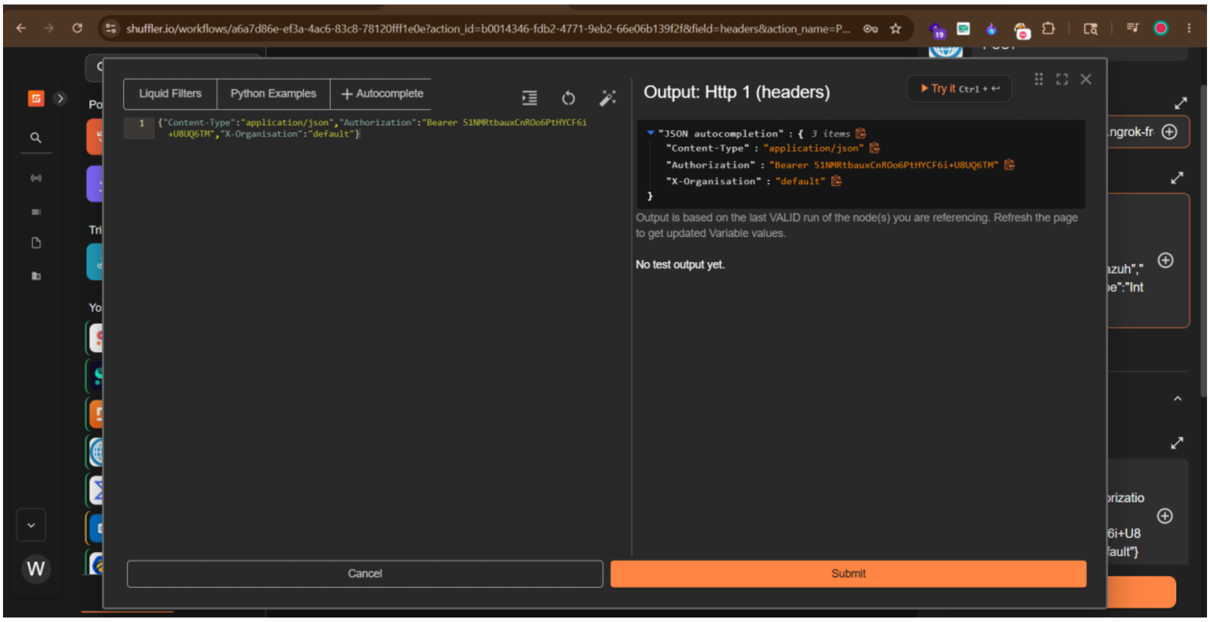
VirusTotal v3 was added to the Shuffle workflow to enrich the SHA256 hash extracted from alerts

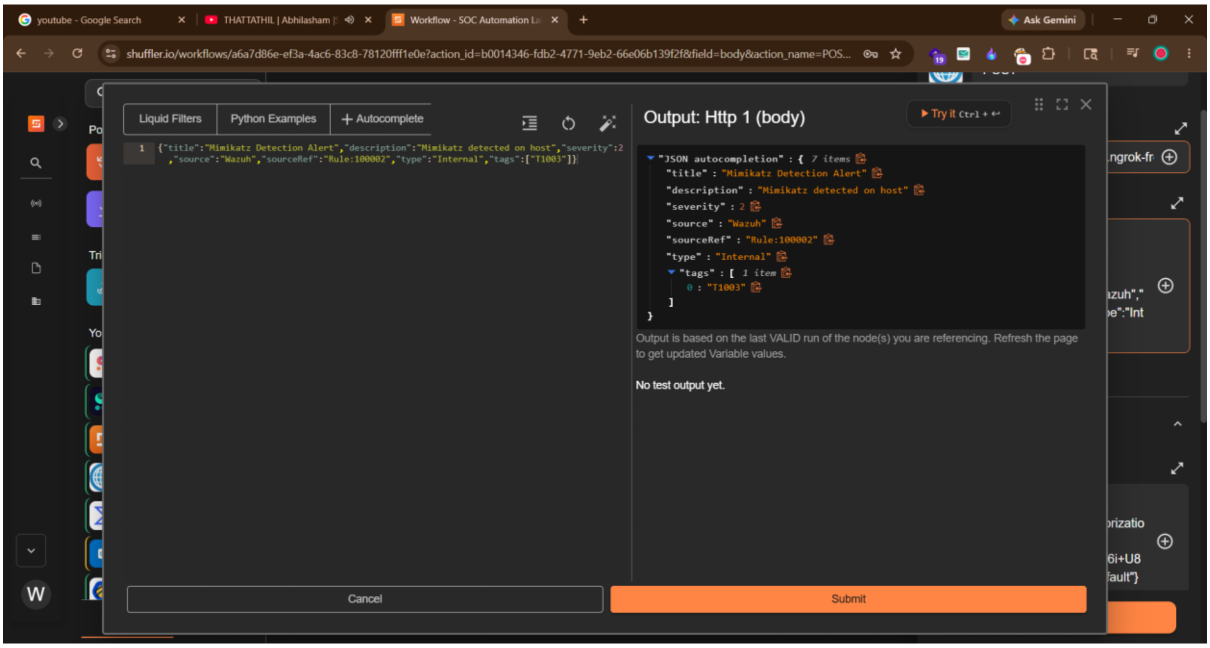


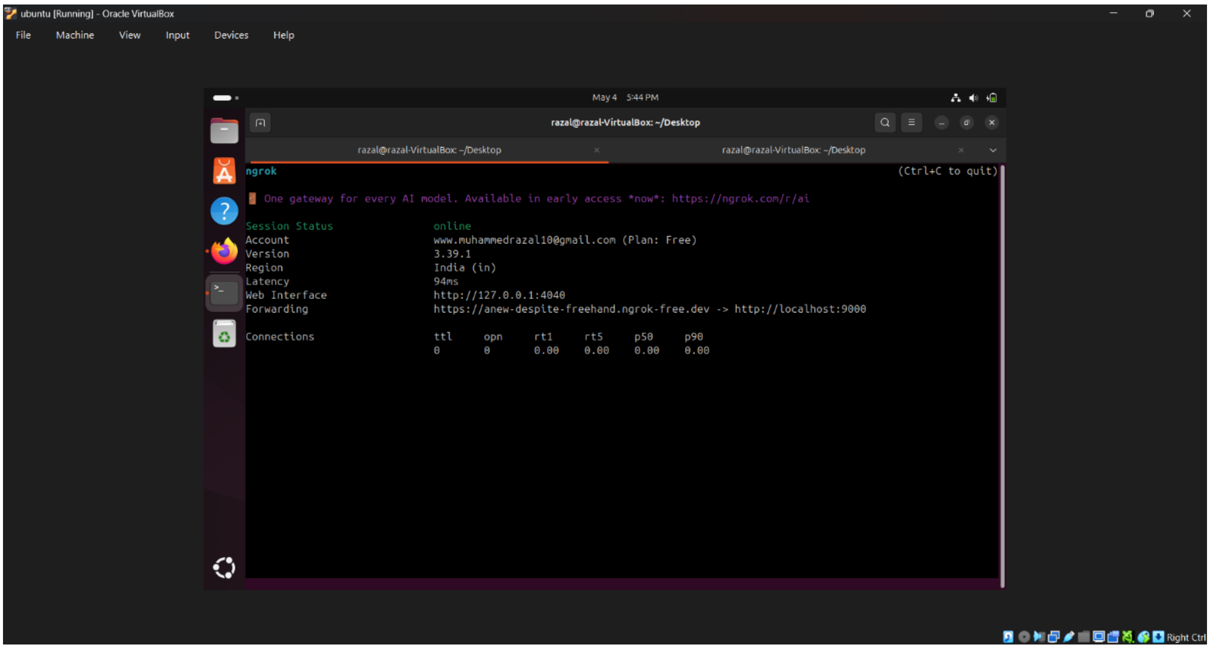


Configuring Shuffle to Work with TheHive

TheHive was added to the workflow with authentication using the org-admin API key and the ngrok public URL



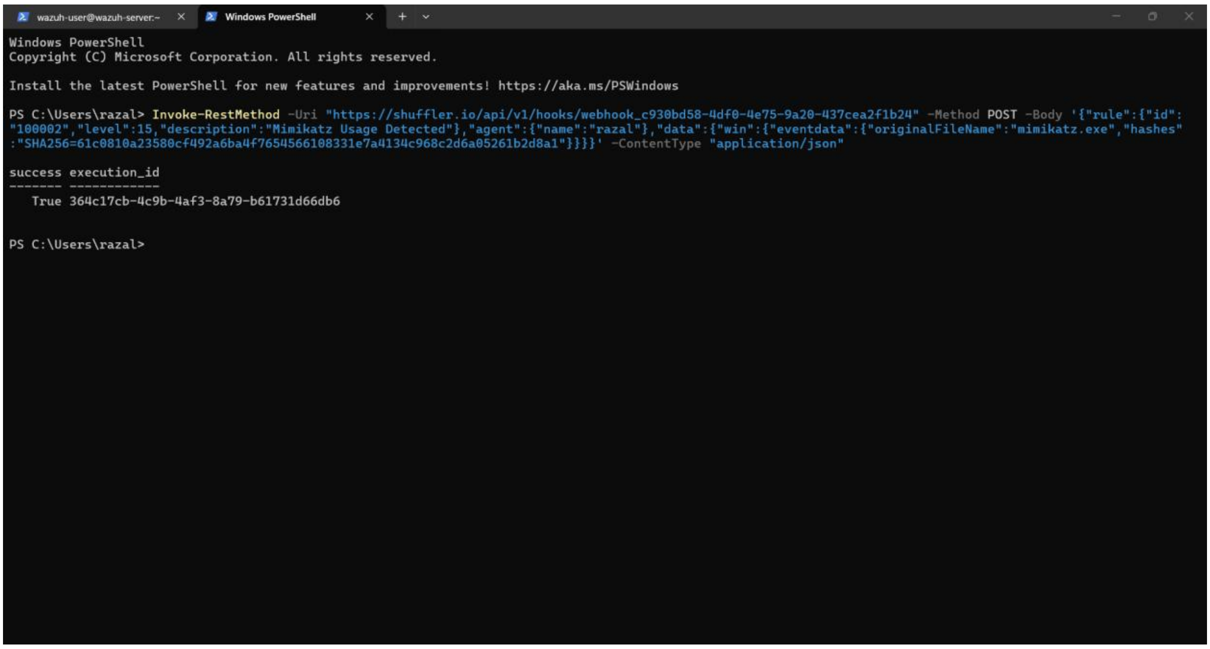

Liquid Filters
Python Examples
Autocomplete
{"title":"Mimikatz Detection Alert","description":"Mimikatz detected on host","severity":2,"source":"Wazuh","sourceRef":"Rule:100002","type":"Internal","tags":["T1003"]}
Output: Http 1 (body)
Try it
"JSON autocompletion" : { 7 items
"title" : "Mimikatz Detection Alert"
"description" : "Mimikatz detected on host"
"severity" : 2
"source" : "Wazuh"
"sourceRef" : "Rule:100002"
"type" : "Internal"
"tags" : [1 item
0 : "T1003"
]
}
Output is based on the last VALID run of the node(s) you are referencing. Refresh the page to get updated Variable values.
No test output yet.
Cancel
Submit


ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
May 4 5:44 PM
razal@razal-VirtualBox: ~/Desktop
ngrok
(Ctrl+C to quit)
One gateway for every AI model. Available in early access *now*: https://ngrok.com/r/ai
Session Status online
Account www.muhammedrazal10@gmail.com (Plan: Free)
Version 3.39.1
Region India (in)
Latency 94ms
Web Interface http://127.0.0.1:4040
Forwarding https://anew-despite-freehand.ngrok-free.dev -> http://localhost:9000
Connections ttl opn rt1 rt5 p50 p90
0 0 0.00 0.00 0.00 0.00

Testing and Results

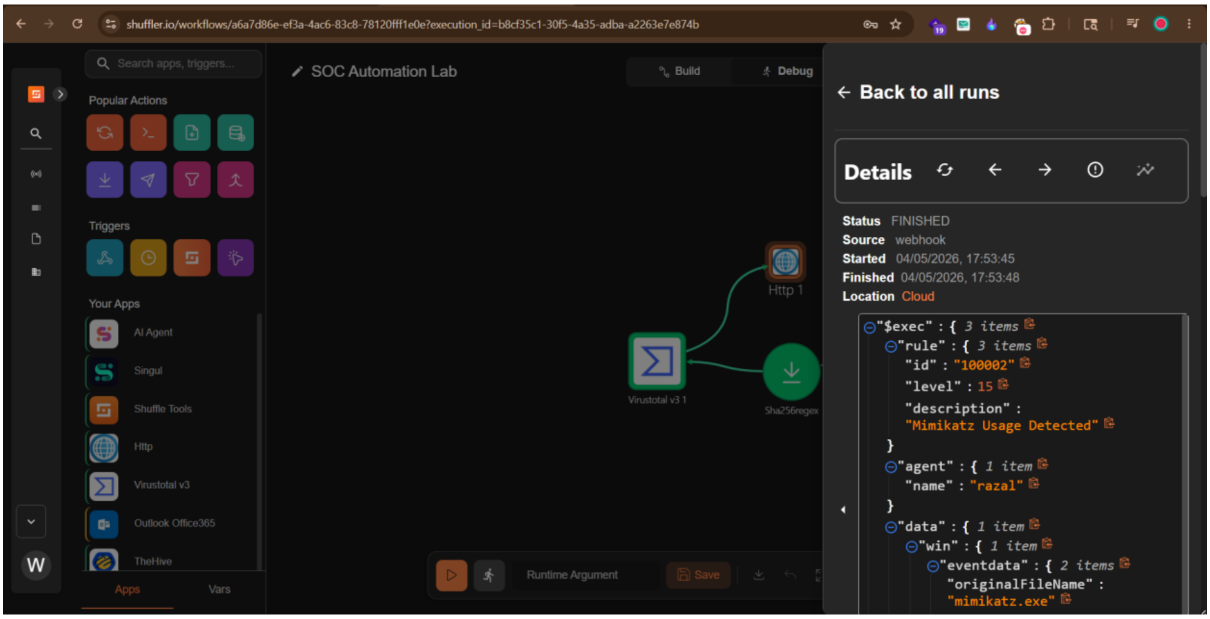
Test Execution

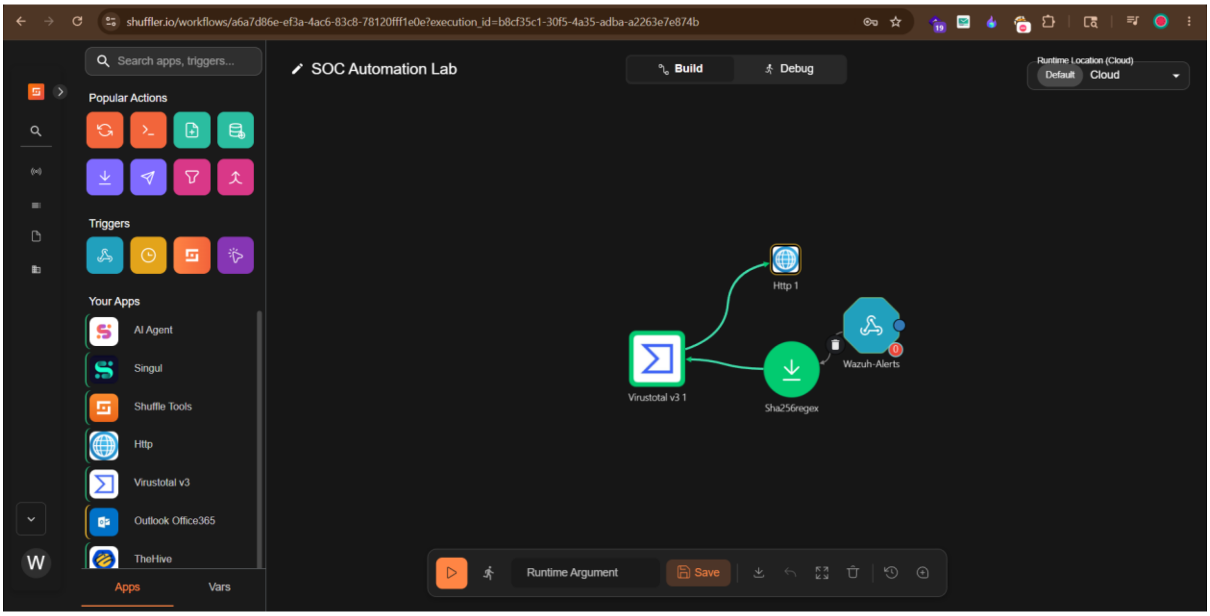
A simulated Mimikatz alert was sent directly to the Shuffle webhook using PowerShell to test the complete pipeline without executing actual malware on the system.



Shuffle Workflow Execution

The Shuffle Debug tab confirmed the workflow executed successfully with Status: FINISHED across all nodes.

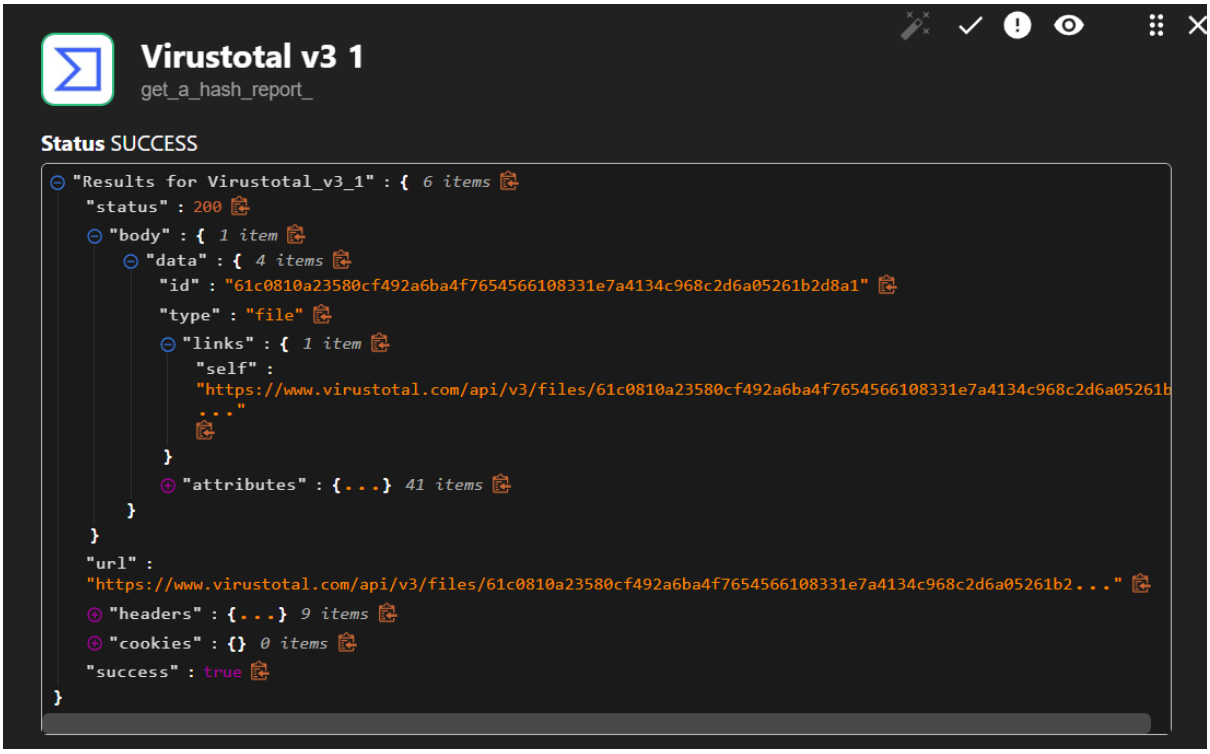




shuffler.io/workflows/a6a7d86e-ef3a-4ac6-83c8-78120fff1e0e?execution_id=b8cf35c1-30f5-4a35-adba-a2263e7e874b
Search apps, triggers...
Popular Actions
Triggers
Your Apps
AI Agent
Singul
Shuffle Tools
Http
Virustotal v3
Outlook Office365
TheHive
Apps
Vars
SOC Automation Lab
Build
Debug
Runtime Location (Cloud)
Default
Cloud
Http 1
Wazuh-Alerts
Virustotal v3 1
Sha256regex
Runtime Argument
Save

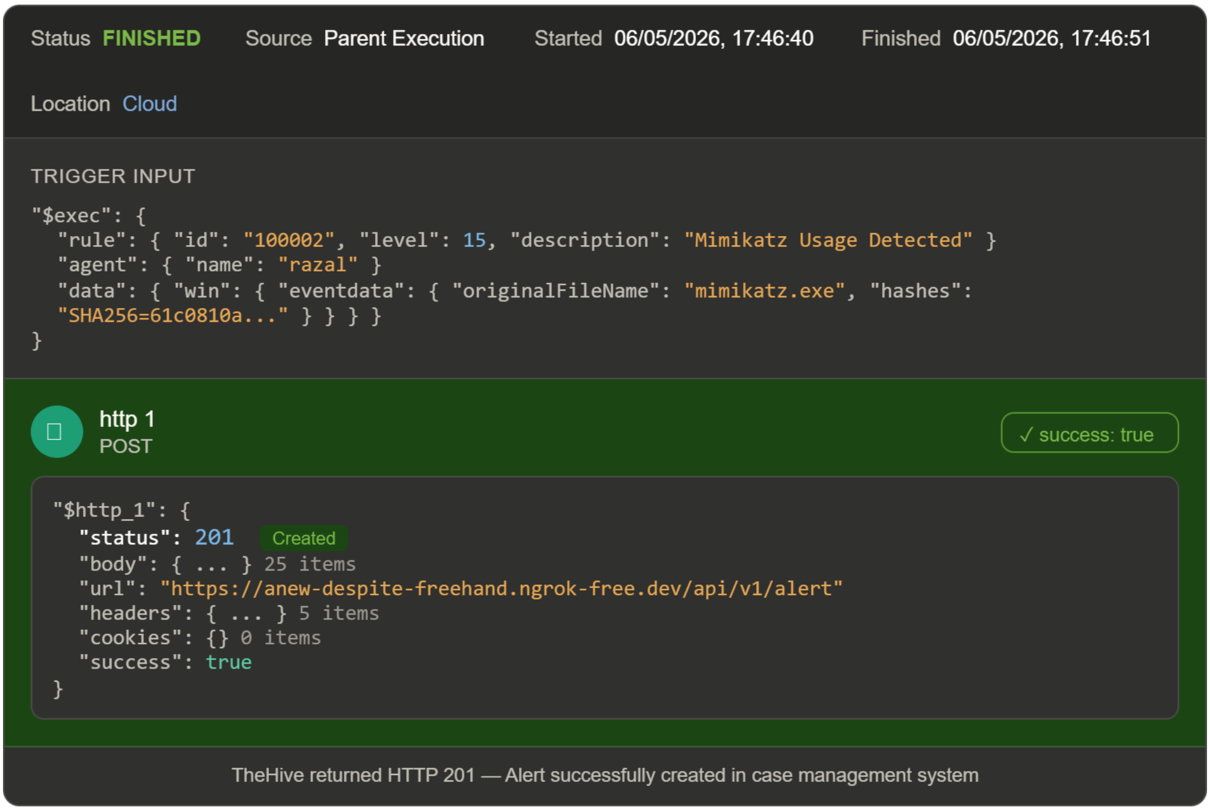
VirusTotal Result

VirusTotal returned a Status 200 response confirming the hash was identified in its database with multiple malicious detections.



TheHive Alert Created

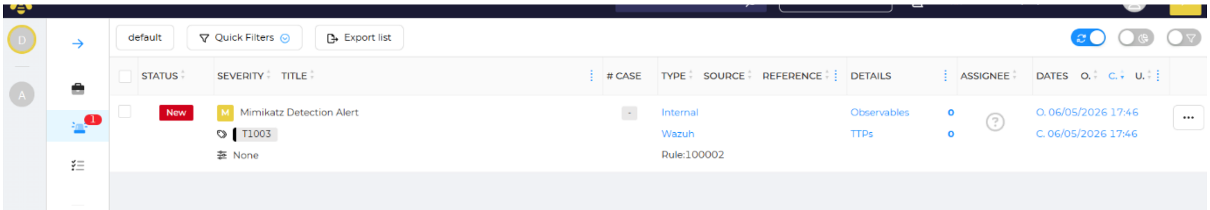
TheHive returned Status 201 confirming the alert was successfully created in the case management system.



Alert in TheHive Dashboard

The Mimikatz Detection Alert appeared in TheHive dashboard with the following confirmed details:

Component	Details
Title	Mimikatz Detection Alert
Status	New
Severity	Medium (2)
Source	Wazuh
Source Reference	Rule:100002
Type	Internal
Tags	T1003
Created	03/05/2026 07:10



Conclusion

This SOC automation project represents a significant stride towards enhancing security operations efficiency and effectiveness. By integrating Wazuh, TheHive, Shuffle, and Sysmon, a robust and automated security ecosystem was successfully created, capable of detecting, responding to, and managing potential threats in real-time.

The project achieved all its primary objectives:

- Wazuh successfully collected and analyzed security events from Windows 11 via Sysmon integration.
- Custom detection rules were created to identify Mimikatz credential dumping attacks.
- Shuffle automated the complete workflow from alert detection through IOC enrichment to case creation.
- VirusTotal successfully enriched IOCs with threat intelligence confirming malicious file hashes.
- TheHive automatically received structured alerts enabling SOC analyst investigation and response.

This SOC automation project serves as a testament to the power of open-source tools and collaborative efforts in building robust and efficient security solutions. It empowers students to gain hands-on experience with cutting-edge security technologies and prepares them for future careers in cybersecurity.